Simplifying Deployment on the Cloud with Heroku

Many developers assume cloud deployment to be tedious, costly, insecure and even sometimes unreliable. However, with the advent of platforms like Heroku, deployment on the cloud is being widely adopted and it is empowering a large number of developers. Let us dive into Heroku and its offerings.

Heroku is a PaaS (Platform as a Service) cloud infrastructure service. Wikipedia defines the Platform as a Service (PaaS) model as a "…category of service models that provide a computing platform *(http://en.wikipedia.org/wiki/Computing_platform)* and a solution stack *(http://en.wikipedia.org/wiki/Solution_stack)* as a service." Simply put, a PaaS offering facilitates application deployment without the need to administer the underlying software and hardware stack. Developers can focus their energies on building applications, which matters most.

There are tons of cloud players competing for developer attention, yet Heroku stands out from the herd. Instead of users purchasing virtualised server instances and managing software installations on them, the Heroku approach is different and innovative—code is pushed into a Git (a distributed version control system) repository, which is then instantly deployed.

With Heroku, hackers get the flexibility of writing applications in the language of their choice, while using the most appropriate database back-end, and flexibly adding the software and hardware infrastructure resources they require. Heroku also subtly pushes developers to follow recommended software development practices such as using version control, isolating dependencies, storing application configurations in environment variables and scaling via concurrency (parallel utilisation of multiple hardware instances). Personally, as a developer, Heroku is my first choice for work or hobby projects; here are some of the reasons why.

1. **Ease and simplicity:** For a stable, robust and scalable deployment, Heroku requires just a basic acquaintance with the platform's terminologies, and a combination of core development and command-line skills.
2. **Trust:** Heroku is built on Amazon Web Services, which is known for its excellent uptime and scalability. Also, Heroku's parent company, *Salesforce.com*, is a trusted enterprise across the world.
3. **Community support and documentation:** With an excellent community backing the platform, and rich detailed documentation for beginners, lots of developers are introduced to Heroku early in their career, and stay loyal to the platform.